CHAPTER

26

Neural Networks (and more!)

Traditional DSP is based on *algorithms*, changing data from one form to another through step-by-step procedures. Most of these techniques also need *parameters* to operate. For example: recursive filters use recursion *coefficients*, feature detection can be implemented by correlation and *thresholds*, an image display depends on the *brightness* and *contrast* settings, etc. Algorithms describe what is to be done, while parameters provide a benchmark to judge the data. The proper selection of parameters is often more important than the algorithm itself. Neural networks take this idea to the extreme by using very simple algorithms, but many highly optimized parameters. This is a revolutionary departure from the traditional mainstays of science and engineering: mathematical logic and theorizing followed by experimentation. Neural networks replace these problem solving strategies with trial & error, pragmatic solutions, and a "this works better than that" methodology. This chapter presents a variety of issues regarding parameter selection in both neural networks and more traditional DSP algorithms.

Target Detection

Scientists and engineers often need to know if a particular object or condition is present. For instance, geophysicists explore the earth for oil, physicians examine patients for disease, astronomers search the universe for extra-terrestrial intelligence, etc. These problems usually involve comparing the acquired data against a threshold. If the threshold is exceeded, the **target** (the object or condition being sought) is deemed present.

For example, suppose you invent a device for detecting cancer in humans. The apparatus is waved over a patient, and a number between 0 and 30 pops up on the video screen. Low numbers correspond to healthy subjects, while high numbers indicate that cancerous tissue is present. You find that the device works quite well, but isn't perfect and occasionally makes an error. The question is: how do you use this system to the benefit of the patient being examined?